SREE MUTHUKUMARASWAMY COLLEGE

DISCIPLINE • KNOWLEDGE • PROGRESS

PROJECT TITLE

Pocket Smart AI

Your smart budget & recommendation assistant

Spend Smart
Live Better

Better Budget Smarter Choices



TEAM LEADER

Tamilselvan.J

MEMBERS

Yogan.P
Akshaya.S
Akshayapriya.K
Bhuveneshwari.A

Small Steps Big Dreams

Smart Money Bright Future

Pocket Smart AI

Your Smart Budget & Recommendation Assistant



- Smart Budget Planning
- Expense Tracking
- Personalized Recommendations
- Savings Tips

Project Introduction

Pocket Smart AI is an AI-powered web application that helps users manage their budgets, track expenses, and get personalized recommendations based on their spending habits and preferences.

Objectives

- To build an AI-powered budget assistant that helps users plan and manage their finances.
- To personalize recommendations based on spending habits and preferences.
- To simplify money management and make it easy to understand.
- To encourage smart spending and healthy financial habits.

Key Features

- **Budget Creation** – Set monthly/weekly budgets with goals.
- **Expense Categorization** – Auto-categorize expenses (e.g., food, travel, shopping, bills).
- **Spending Analysis** – View charts and insights on spending patterns.
- **Product/Service Recommendations** – Get relevant suggestions based on your spending and interests.
- **Savings Alerts** – Receive notifications for overspending and saving opportunities.

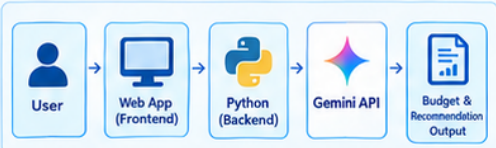
How It Works

User Input (Goals, preferences, expenses) → **Python Backend** (Processes data) → **Gemini AI** (Analyzes & generates insights) → **Personalized Budget & Recommendations** → **User Tracks Spending** (Updates & improves)

Technology Used

- HTML / CSS / JavaScript (or Streamlit) – Frontend
- Python – Backend
- Google Gemini AI – AI model
- pandas – Data analysis
- google-generativeai – API library

System Design



Gemini AI Integration

- We use Google's Gemini Pro model to generate personalized budget plans and recommendations.
- The user's input (goals, spending habits, preferences) is sent to Gemini with a structured prompt.
- Gemini returns a detailed and customized plan in natural language format.

gemini_integration.py

```python
import google.generativeai as genai

genai.configure(api_key="YOUR_GEMINI_API_KEY")

model = genai.GenerativeModel("gemini-pro")

def get_recommendations(prompt: str) -> str:
    response = model.generate_content(prompt)
    return response.text

# Example usage
prompt = "Create a personalized budget plan and
recommendations based on this user data."
print(get_recommendations(prompt))
```

Gemini helps us get smart, accurate and personalized budget suggestions.

Coding (Python + HTML)

app.py (Python - Streamlit)

```python
import streamlit as st
import pandas as pd
import google.generativeai as genai

genai.configure(api_key="YOUR_GEMINI_API_KEY")
model = genai.GenerativeModel("gemini-pro")

def analyze_expenses(df):
    total = df['amount'].sum()
    by_cat = df.groupby('category')['amount'].sum()
    return total, by_cat

st.title("Pocket Smart AI")
st.sidebar.title("Menu")
page = st.sidebar.radio("Go to",
    ["Home", "Budget Planner", "Expenses",
     "Recommendations", "Settings"])

if page == "Home":
    st.write("Welcome to Pocket Smart AI!")
    # more app logic...
```

index.html (Frontend - Simple UI)

```html
<!DOCTYPE html>
<html>
<head>
    <title>Pocket Smart AI</title>
    <link rel="stylesheet" href="style.css">
    <meta charset="UTF-8">
</head>
<body>
    <div class="container">
        <h1>Pocket Smart AI</h1>
        <div class="sidebar">
            <a href="#" class="active">Home</a>
            <a href="#">Budget Planner</a>
            <a href="#">Expenses</a>
            <a href="#">Recommendations</a>
            <a href="#">Settings</a>
        </div>
        <div class="content">
            <h2>Your Financial Companion</h2>
        </div>
    </div>
    <script src="app.js"></script>
</body>
</html>
```

Sample Output

Budget Summary

Monthly Budget	₹15,000
Total Spent	₹9,250
Remaining	₹5,750

Expense Analysis



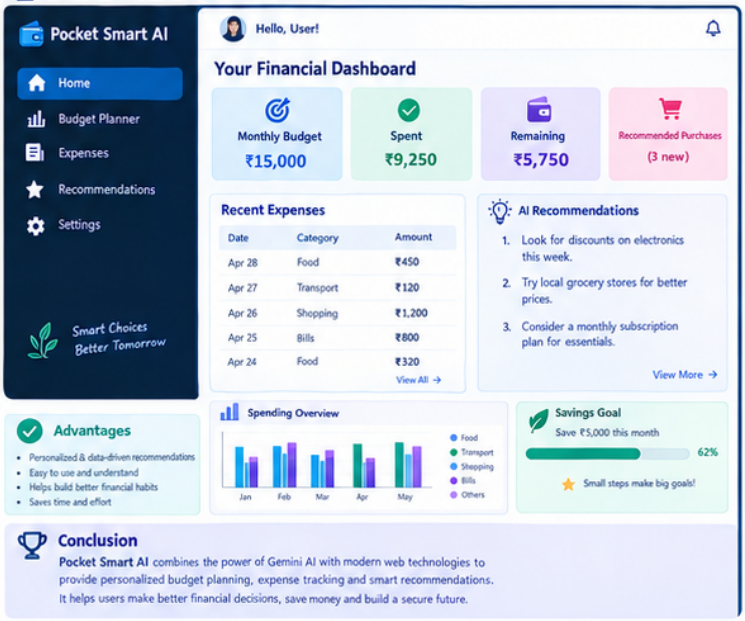
Recommendations

- Buy essentials online (better deals)
- Try budget-friendly brands
- Avoid non-essential purchases

Savings Tips

- Save at least 20% monthly
- Avoid impulse buying
- Use cashback & rewards

Final Output (Web Application Screenshot)



Conclusion

Pocket Smart AI combines the power of Gemini AI with modern web technologies to provide personalized budget planning, expense tracking and smart recommendations. It helps users make better financial decisions, save money and build a secure future.

Project Introduction

1. Executive Summary

PocketSmart AI is an AI-powered personal finance and budgeting assistant. It helps users track expenses, create budgets, understand spending patterns, and receive personalized recommendations. The application combines a simple web interface, Python-based processing, data analysis, and Google Gemini AI.

2. Problem Statement

Many users find it difficult to monitor daily expenses, control unnecessary spending, and maintain consistent savings. Traditional budgeting methods can be time-consuming and may not provide personalized guidance.

3. Objectives

- Build an AI-powered budget assistant that helps users plan and manage finances.
- Personalize recommendations using spending habits, goals, and preferences.
- Simplify money management through a clear interface.
- Encourage smart spending and healthy financial habits.

4. Scope of the Project

The system covers expense tracking, category-wise analysis, budget planning, savings guidance, and AI-generated recommendations. Future versions can connect to financial data sources and provide automated alerts.

Key Features & System Architecture

1. Key Features

- **Budget Creation:** Set monthly or weekly budgets with financial goals.
- **Expense Tracking:** Record and organize daily transactions.
- **Expense Categorization:** Classify spending into food, travel, shopping, bills, and other categories.
- **Spending Analysis:** Calculate totals and identify spending patterns.
- **Personalized Recommendations:** Generate suggestions based on user data.
- **Savings Tips:** Highlight possible saving opportunities.

2. System Design

User → Web App / Frontend → Python Backend → Gemini AI → Budget & Recommendation Output

3. Technology Stack

- Frontend: HTML, CSS, JavaScript / Streamlit
- Backend: Python
- Data Analysis: Pandas
- AI: Google Gemini
- API Library: google-generativeai

4. Data Flow

The user enters income, expenses, preferences, and goals. The backend validates and processes the information. Spending totals are calculated, a structured prompt is prepared, Gemini generates recommendations, and the final results are displayed on the dashboard.

AI Integration & Backend Implementation

1. Gemini AI Integration

Google Gemini can be used to turn structured financial information into clear, natural-language recommendations. The application sends the user's goals, spending habits, and budget information through a structured prompt.

2. Python Backend – Sample Code

```
import pandas as pd
import google.generativeai as genai

genai.configure(api_key="YOUR_GEMINI_API_KEY")
model = genai.GenerativeModel("gemini-pro")

def analyze_expenses(df):
total = df["amount"].sum()
by_category = df.groupby("category")["amount"].sum()
return total, by_category

def get_recommendations(prompt):
response = model.generate_content(prompt)
return response.text
```

3. Recommendation Logic

The application can compare spending with the user's budget limits, identify high-spending categories, calculate remaining funds, and ask the AI model to produce practical suggestions.

4. Example Recommendation Prompt

“Create a personalized budget plan based on the user’s monthly income, current expenses, savings goal, and spending preferences. Suggest practical ways to reduce unnecessary spending.”

5. Important Security Note

API keys should be stored securely using environment variables or a protected secrets manager. They should not be exposed in frontend code or publicly shared project files.

Final Output, Advantages & Future Scope

1. Sample Financial Dashboard

Monthly Budget: ■15,000 **Total Spent:** ■9,250 **Remaining:** ■5,750

Example AI suggestions: compare prices before shopping, reduce non-essential purchases, set a monthly savings target, and avoid impulse purchases.

2. Advantages

- Easy-to-use budgeting workflow.
- Personalized and data-driven recommendations.
- Helps users understand spending patterns.
- Encourages consistent saving habits.
- Can be expanded with additional AI and financial features.

3. Future Enhancements

- Automatic transaction categorization using machine learning and NLP.
- Bank / Open Banking integration where supported.
- Push notifications for budget limits and saving opportunities.
- Savings-goal tracking and progress dashboards.
- Advanced monthly and yearly spending charts.
- More personalized AI recommendations based on long-term patterns.

4. Conclusion

PocketSmart AI combines modern web technologies, data analysis, and generative AI to create a practical budgeting assistant. By organizing expenses, analyzing spending, and generating personalized recommendations, the project demonstrates how AI can be applied to everyday financial planning.

“Spend Smart • Save More • Live Better”